

WHEN DO REUSABLE TEMPORAL PROCEDURES ACTUALLY REPAIR AGENTS?

Anonymous authors

Paper under double-blind review

ABSTRACT

Agents that revisit evolving data can repeat the same procedural mistake even as the values change: using evidence released after a cutoff, confusing publication time with reporting time, or failing to ground a numerical change in contemporaneous documents. We study an identification problem rather than a skill-learning problem: when does inserting a reusable temporal procedure actually repair such a recurring failure, and what part of the intervention deserves credit? Our base design compares a targeted procedure (T), a complexity-matched target-blind helper (G), and the original agent (N); a binding repair requires T to beat both G and N. This extra anchor changes an observed verdict: on one DeepSeek cutoff cell, $T=100\%$ and $G=72\%$, but $N=100\%$, so the apparent +28-point two-arm gain is not a repair. We then strengthen identification with two prospectively frozen controls. A no-op helper G_0 is behaviorally neutral relative to fresh N on independent DeepSeek and Kimi tracks (90% equivalence intervals contained in ± 10 points). An operation-matched retrieval treatment R runs the same frozen temporal operation but materializes its output in retrieved context rather than callable skill state. Across 18 pre-frozen DeepSeek load-bearing endpoints, $T-R$ is exactly 0.0 points with a 90% endpoint-bootstrap interval of $[-5.6, 5.6]$; three held-out Kimi alignment endpoints are also identical under T and R across five fresh repeats. Thus the evidence supports conditional *operation-level* temporal bottlenecks in non-ceiling regimes, but not a material advantage for the callable skill container. The strongest held-out cutoff cell still reaches 100% under T versus 47.5% under N, while cross-domain grounding and a fresh Kimi grounding recheck remain null. The resulting claim is narrower and better identified: some recurring temporal operations are binding repairs, and retrieval/context construction can implement the same repair at the tested resolution.

1 INTRODUCTION

Suppose an agent is asked: *What was the latest value known by the end of April?* A May release is highly relevant to the topic, easy to retrieve, and wrong for the question. On the next period the numbers change, but the required operation does not: filter evidence by the information cutoff. Similar recurring operations arise when an agent must distinguish a publication date from the reporting period it describes, or ground a numerical change in documentary context that was actually available at the time. These are not one-off facts to memorize. They are procedures that should remain useful as institutional releases evolve.

Reusable agent skills are one appealing way to persist such procedures, but the scientific object is the operation rather than the container that stores it. Demonstrating that the operation helps is subtler than comparing a targeted helper with a generic one. If generic assistance distracts the model, then the targeted helper can win while doing no better than the original agent. In that case a two-arm experiment reports a positive “skill effect” even though nothing has been repaired. This is not hypothetical in our data: for one frozen cutoff cell, the targeted procedure scores 100%, the matched generic helper 72%, and the no-skill agent 100%. The targeted-minus-generic contrast is +28 points, but the correct repair verdict is null.

This example separates three questions that are often conflated. *Repair value* asks whether T improves on the original agent N. *Operation specificity* asks whether T beats structured assistance that lacks

the target operation. We test this twice: first against the complexity-matched target-blind G, then against a separately validated no-op helper G_0 whose behavior is empirically equivalent to fresh N within a prospectively frozen ± 10 -point margin. *Container value* asks whether the same operation works better as callable skill state than as an operation-matched retrieval/context treatment R. We execute this comparison as well: R runs the identical frozen operation on the identical candidate evidence and places the same operation output in retrieved context, changing only the integration surface. This completed ladder lets us separate operation value from generic-helper degradation and from callable-container value.

Our question is therefore: *when does a reusable temporal procedure become a binding repair for a recurring failure?* We call such a case an *intervention-defined causal bottleneck*. For the same frozen endpoint we evaluate three arms: a mechanism-specific targeted procedure (T), a complexity-matched target-blind helper (G), and the original no-skill agent (N). A model-task cell qualifies only when T improves over both G and N under the family outcome frozen before model results are opened. The term *bottleneck* refers only to an operation whose insertion is binding under this executed intervention; it does not claim causal exclusivity, behavioral neutrality of G, or necessity of the callable-skill container.

This definition yields three falsifiable predictions. First, $T > G$ without $T > N$ must be rejected as generic-helper degradation rather than counted as repair. Second, a real procedure bottleneck should be visible only where the original model has headroom; cells already at no-skill ceiling should show no identifiable gain. Third, if the reusable operation rather than endpoint-specific refitting is responsible, byte-identical procedure code should retain its direction on compatible held-out releases, while transfer to an incompatible domain may legitimately be null.

The question is deliberately narrower than recent work on skill learning. Paired trace auditing, no-skill attribution, randomized skill masking, continual skill evaluation, and skill compilation already establish that reusable skills can change agent behavior and that their effects are heterogeneous (Zhou et al., 2026; Dong et al., 2026; Wang et al., 2026b; Guan et al., 2026; Wei et al., 2026; Zheng et al., 2026; Xu et al., 2026). Likewise, RAG systems and temporal retrieval methods can implement time-aware evidence selection upstream (Asai et al., 2024; Yan et al., 2024; Han et al., 2025; Wang et al., 2026a). We therefore do not claim that reusable procedures, temporal filtering, or paired skill causality are new. Our object is the stricter *diagnosis*: whether a predeclared recurring failure is actually repaired by a particular reusable operation rather than merely looking better against a weakened reference.

We study this object using source-native replay built from first-party U.S. Bureau of Economic Analysis (BEA), NOAA Climate Prediction Center, and Energy Information Administration (EIA) release histories (U.S. Bureau of Economic Analysis, 2026; NOAA Climate Prediction Center, 2026; U.S. Energy Information Administration, 2026). The complete evidence ledger now contains 35 frozen endpoints and 2,056 retained runtime-valid rows, including fresh N/ G_0 /T and T/R identification runs. The initial BEA/NOAA replay preserves three mechanism families motivated by TimeSage-EV—cutoff validity, release alignment, and exogenous grounding—while remaining a separate, content-addressed experimental object (Yao et al., 2026). A post-review EIA expansion adds 12 cutoff endpoints using the same targeted implementation byte-for-byte. Repeats test execution robustness; endpoints, not repeats, are the independent scientific units.

The results follow the identification ladder rather than a universal “skills help” story. The no-skill anchor first overturns the apparent +28-point DeepSeek cutoff repair described above. In non-ceiling cells, targeted procedures still beat the original agent and target-blind help: DeepSeek documentary grounding reaches 60% versus 20%/20%, and held-out EIA cutoff filtering reaches 100% versus 70%/47.5%, with an endpoint sign test of $p = 0.0156$ for T–N. The stronger controls then sharpen what these gains mean. G_0 is neutral relative to fresh N on both DeepSeek and Kimi, while T and operation-matched R are indistinguishable on all three DeepSeek load-bearing cells at the frozen resolution (pooled T–R = 0.0, 90% CI $[-5.6, 5.6]$ points). Kimi held-out alignment likewise gives T=R=100% across five fresh repeats per endpoint. Conversely, a fresh Kimi same-domain grounding recheck does not preserve its earlier apparent gain. The evidence therefore supports conditional temporal *operations*, not a universal skill effect or a privileged callable container.

Contributions. Our contribution is an identification contract for reusable-procedure repair, not a new skill optimizer. We (i) define a base three-arm bottleneck test that requires improvement over

both the original agent and target-blind assistance; (ii) show an observed two-arm false positive in which adding N changes the scientific verdict; (iii) prospectively validate a behavior-neutral no-op control G_0 , separating operation value from generic-helper degradation; and (iv) execute an operation-matched retrieval/context treatment R, showing that the measured benefit is attributable to the temporal operation rather than to callable-skill packaging. Unchanged-operation held-out replay then exposes both non-ceiling positives and explicit null boundaries. The resulting claim is intentionally conditional on operation $\times$ model $\times$ task regime and explicitly compatible with temporal retrieval implementations.

2 FROM RECURRING FAILURE TO A PROCEDURE BOTTLENECK

The intuition is to treat a reusable procedure as an intervention, not as a label attached to a successful run. The original agent must first provide the reference behavior: if it already succeeds, a targeted helper cannot be credited with repairing that endpoint. A target-blind helper then tests whether the gain is specific to the nominated operation rather than to receiving some structured assistance. Only after both comparisons are visible do we ask whether the same unchanged operation retains its direction on future compatible endpoints.

Let endpoint i specify a question x_i , a cutoff-valid evidence package E_i , a model configuration M_i , and a family-specific binary success rule $Y_i \in \{0, 1\}$. We intervene only on the reusable procedure exposed to the frozen harness. The three executed conditions are N (no experimental helper), G (a complexity-matched target-blind helper), and T (the mechanism-specific targeted procedure). All other endpoint inputs and scoring rules are held fixed.

For a set of frozen endpoints $\mathcal{I}$ within one model-task regime, we report two empirical contrasts,

$$\hat{\Delta}_{T-N} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (\bar{Y}_i(T) - \bar{Y}_i(N)), \quad \hat{\Delta}_{T-G} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (\bar{Y}_i(T) - \bar{Y}_i(G)), \quad (1)$$

where $\bar{Y}_i(a)$ averages repeated executions of the same endpoint under arm a . Repeats characterize execution robustness and do not increase the number of independent endpoint units. The two contrasts have different jobs: $\hat{\Delta}_{T-N}$ asks whether inserting the procedure changes the original agent, whereas $\hat{\Delta}_{T-G}$ asks whether the targeted operation outperforms the executed target-blind helper. Neither contrast establishes that G is behaviorally neutral or that the callable-skill container itself is necessary.

Definition 1 (intervention-defined procedure bottleneck). A targeted procedure is a *binding repair* under the base three-arm intervention for a frozen model-task regime only when it improves the registered family outcome over both the original agent and matched generic assistance, i.e., when the endpoint evidence supports $\hat{\Delta}_{T-N} > 0$ and $\hat{\Delta}_{T-G} > 0$. A positive targeted-minus-generic contrast with $\hat{\Delta}_{T-N} \leq 0$ is explicitly classified as *no repair*. This base definition identifies the value of inserting the operation; stronger controls then determine whether that value survives behaviorally neutral extra computation and whether the callable container itself matters.

Stronger identification layers. We prospectively freeze and execute two additional treatments. G_0 is a no-op helper that uses the same helper-execution surface as T but returns an empty object and reads neither the evidence package nor task context. It is admitted as a neutral control only if fresh G_0 and fresh N are equivalent within a predeclared ± 10 -point margin. R executes the *same frozen temporal operation* as T on the same candidates, but places the identical operation output in the retrieved evidence context and exposes no callable helper. Thus

$$\hat{\Delta}_{T-G_0} = \frac{1}{|\mathcal{I}|} \sum_i (\bar{Y}_i(T) - \bar{Y}_i(G_0)), \quad \hat{\Delta}_{T-R} = \frac{1}{|\mathcal{I}|} \sum_i (\bar{Y}_i(T) - \bar{Y}_i(R)) \quad (2)$$

separate operation specificity from callable-container value. A $T > G_0$ residual supports value beyond neutral extra computation. By contrast, $T \approx R$ supports the operation while rejecting the stronger claim that callable skill state is necessary.

The design targets three failure modes that recur as release content changes.

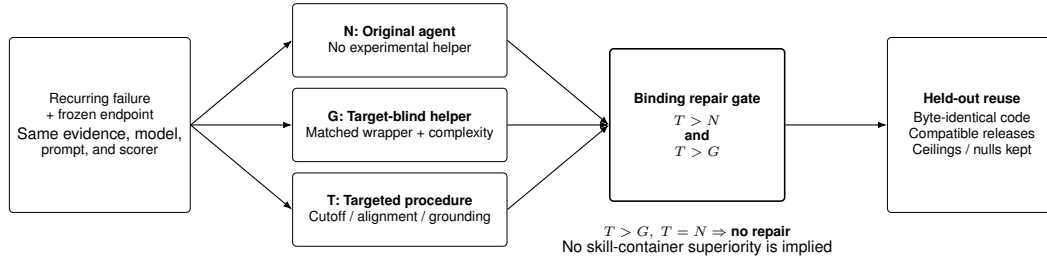


Figure 1: Three-arm procedure-bottleneck test. The same frozen endpoint is evaluated as the original no-skill agent (N), with a complexity-matched target-blind helper (G), and with the targeted reusable procedure (T). A binding repair requires T to beat both N and G; unchanged-code held-out replay separately tests reuse.

2.1 CUTOFF VALIDITY

The cutoff procedure receives a query date and candidate evidence timestamps and returns only evidence admissible by that date. It addresses future leakage: a later release may be highly relevant yet invalid for an earlier information set. The matched generic helper can validate evidence-package structure but is forbidden from comparing release dates with the cutoff.

2.2 RELEASE ALIGNMENT

The alignment procedure maps a source release to the reporting period represented by its content and verifies the selected time-series row against that period. It addresses cases in which publication time and reporting time differ. The matched generic helper can inspect document structure but cannot perform the date-to-period mapping.

2.3 EXOGENOUS GROUNDING

The grounding procedure links a numerical change to first-party documentary evidence available before the cutoff. It operates over the full frozen candidate-span region rather than a gold-only pool and returns evidence tied to the requested attribution slots. Its generic helper exposes document structure without the mechanism-specific contextual event selector.

No targeted procedure contains scenario-specific answers, gold labels, or endpoint-specific fitted values. Source hashes are frozen before execution, and unchanged implementations are reused on held-out periods. This gives three direct tests of the bottleneck interpretation: a two-arm false positive must be rejected by N; gains should disappear at no-skill ceiling; and compatible held-out releases should preserve direction without rewriting the procedure.

3 RELATED WORK

Causal skill attribution. Paired trace auditing, matched/no-skill comparisons, and randomized skill masking already show that exposing an agent to a skill can causally change behavior and that per-skill effects are heterogeneous (Zhou et al., 2026; Dong et al., 2026; Wang et al., 2026b). We therefore do not claim paired skill causality, no-skill attribution, or heterogeneous skill effects as new. Our residual question is stricter: for a *predeclared recurring failure*, does one targeted operation repair the original agent while also outperforming matched generic assistance? The observed $T > G$ but $T = N$ cutoff cell shows why the second comparison changes the scientific verdict rather than merely adding another baseline.

Skill evolution and reusable procedures. ContinualSkillBench, Evo-Harness, SkillProx, HyperSkill, and Anything2Skill study continual skill maintenance, compilation, optimization, structured skill memory, and conversion of external knowledge into reusable procedures (Guan et al., 2026; Wei et al., 2026; Zheng et al., 2026; Xu et al., 2026; Pan et al., 2026). These works make it untenable to

claim that procedure reuse, cross-task skill accumulation, or external-knowledge-to-skill compilation is itself novel. We instead treat skill acquisition as upstream of our object: once a reusable procedure exists, our three-arm contract asks whether it is a *binding repair* for the recurring failure that motivated its persistence, and unchanged-code replay tests where that diagnosis transfers or disappears at ceiling.

Temporal retrieval and generic assistance. Selective/corrective RAG and temporal retrieval systems already address irrelevant evidence, time-sensitive indexing, and explicit temporal constraints upstream of generation (Asai et al., 2024; Yan et al., 2024; Zhang et al., 2025; Abdallah et al., 2026; Han et al., 2025; Wang et al., 2026a; Liu et al., 2026). A useful cutoff or alignment operation therefore need not be packaged as a callable skill. We test this directly rather than leaving it as a limitation. Our R treatment runs the same frozen T operation on the same candidate evidence and places the identical output in retrieved context, while T exposes it through callable state. On the 18 pre-frozen DeepSeek load-bearing endpoints T-R is 0.0 points with a 90% interval of $[-5.6, 5.6]$, and the secondary Kimi alignment check is also exactly tied. We therefore explicitly concede that temporal retrieval/context construction is an equally effective integration surface at the tested resolution. Likewise, G_0 shows that the target-blind G deficits are not an unavoidable cost of adding a helper. The residual scientific object is consequently *procedure-bottleneck identification*: which temporal operation is binding in which model-task regime, not a new temporal-retrieval algorithm and not evidence that skill packaging itself is necessary.

4 TARGETED PROCEDURE INTERVENTION

Each frozen endpoint is replayed under three conditions while its evidence package, model configuration, prompt harness, evaluator, and period remain fixed. The intervention is intentionally simple: compare the original agent with the same agent receiving either a target-blind helper or the one reusable operation predicted by the registered failure mechanism.

Design requirements. The base comparison must satisfy three requirements. First, the targeted arm must improve over the original no-skill agent; otherwise there is no repaired failure. Second, it must also improve over a helper with the same callable surface and matched implementation complexity; otherwise the effect may reflect generic assistance rather than the target operation. Third, the endpoint outcome and transfer split must be frozen before the corresponding model outcomes are opened, so neither the mechanism label nor the success criterion can move to fit the result. We then add two stronger post-review requirements without changing the registered family outcome: G_0 must first pass a prospective neutrality test against fresh N, and R must expose the same operation output and candidate evidence as T while changing only the integration surface.

The targeted condition injects the registered mechanism-specific procedure. The generic condition injects a frozen target-blind helper under the same callable wrapper. The no-skill condition adds no experimental helper. The scientific unit is the frozen endpoint, not a repeat. We evaluate the two contrasts from Definition 1 jointly: targeted minus no skill asks whether the operation repairs the original agent, while targeted minus generic asks whether that repair exceeds generic helper/code assistance. Neither contrast is sufficient alone.

4.1 MECHANISM CONTRACTS

The cutoff procedure filters candidate evidence by the target cutoff. Its generic control validates package structure without using timestamps. The release-alignment procedure normalizes the reporting period represented by a source release. Its generic control inspects document structure without performing date-to-period mapping. The grounding procedure enriches candidate documentary spans using reusable attribution and outlook cues. Its control can expose document structure but has no contextual event selector.

No targeted procedure contains scenario-specific values, dates, labels, or expected answers. Procedure source hashes are frozen before execution. The targeted and generic source implementations are matched exactly in lexical-token count and AST-node count for each family. This generic control is deliberately *target-blind*, not assumed behaviorally neutral: it controls the callable wrapper and implementation-complexity surface while withholding the temporal operation under test. The no-skill

arm is therefore essential for detecting harm introduced by the particular generic helper. G_0 addresses the remaining concern directly: it executes one helper call but is information-empty, reading neither package nor context and returning only an empty object. Its neutrality is tested empirically rather than assumed.

4.2 RESOURCE PARITY AND IDENTIFICATION LAYERS

The experiment measures controlled procedure exposure, not autonomous skill selection. Under the frozen harness, T, G, and G_0 each execute exactly one assigned helper operation before generation; N executes no experimental helper. R instead executes the same targeted temporal operation during context construction and exposes no callable helper. Thus T-G matches callable surface and source complexity, G_0 tests whether extra helper computation itself perturbs the agent, and T-R holds operation content fixed while changing only where that content enters the prompt. We make no latency or provider-cost superiority claim.

The completed identification ladder has four distinct contrasts. T-N asks whether the operation repairs the original agent. T-G asks whether it beats the executed target-blind helper. T- G_0 asks whether the operation retains value beyond behaviorally neutral extra computation, after G_0 has independently passed its N-equivalence gate. T-R asks whether callable skill state adds value beyond an operation-matched retrieval/context implementation. For R, the original candidate evidence is preserved and the exact T operation output is added as `retrieval_operation_output`; a zero-call audit verifies content equality and that no candidate evidence is removed or reordered. This is intentionally a strongest same-information baseline, not a weaker temporal-RAG surrogate.

4.3 FROZEN FAMILY OUTCOMES AND SCORER TIMELINE

Before any source-native model outcome is opened, the failure-family contracts define what constitutes success. A cutoff success requires a nonempty set of cited evidence and a selected latest reference that are all available by the target cutoff. A release-alignment success requires the correct reporting-period identity together with the aligned headline value. A grounding success requires the requested evidence slots to be filled by frozen first-party sentence IDs in the correct slot order.

An auxiliary full-task conjunctive scorer is preserved only as sensitivity analysis because it adds orthogonal output fields and exact surface-form requirements beyond the registered mechanism outcome. The family definitions frozen before execution remain primary; Appendix A records the scorer timeline and representation audit.

The base decision rule follows Definition 1: $T > G$ with $T \leq N$ is no repair, while $T > N$ without $T > G$ cannot isolate the targeted operation from generic assistance. The stronger controls are interpreted asymmetrically. G_0 can strengthen operation specificity only after its fresh G_0 -N 90% interval lies wholly inside the predeclared ± 10 -point margin. For container value, a new callable-skill claim requires a prospectively resolved T-R residual of at least 10 points with the frozen endpoint-level criterion. If the 90% T-R interval instead rules out a > 10 -point advantage, we retain only the operation-level claim and explicitly concede retrieval/context parity.

4.4 HELD-OUT TRANSFER AND EVIDENCE LAYERS

The initial replay freezes twelve pre-April endpoints as C3-R and eleven April-May endpoints as held-out C4-R; same-domain and compatible cross-domain labels are fixed before outcomes, and procedure source is unchanged across the split. A separate EIA validation freezes four earlier and eight held-out energy endpoints while reusing T1/G1 byte-identically. DeepSeek is the registered primary track, Kimi a provenance-clean replication selected by a pre-outcome support rule, and the mini-model layer an exploratory capacity stress. Full execution chronology, the failed local-secondary gate, and row-level provenance bindings are reported in Appendix A.

5 EXPERIMENTAL SETUP: SOURCE-NATIVE REPLAY

TimeSage-EV motivates the three recurring mechanism families—cutoff validity, release alignment, and exogenous grounding—and exposes persistent skills to later periods (Yao et al., 2026). Our

experiment is not an exact rerun of its unavailable evaluated snapshot. Instead, we reconstruct content-addressed endpoints directly from first-party institutional release histories and intervene on the same reusable operations.

5.1 FROZEN ENDPOINTS AND SCORERS

The initial replay contains 23 endpoints from BEA GDP releases and NOAA-CPC ENSO Diagnostic Discussions (U.S. Bureau of Economic Analysis, 2026; NOAA Climate Prediction Center, 2026): twelve pre-April endpoints form C3-R and eleven April–May endpoints form held-out C4-R. Raw pages are SHA-256 bound, evidence identifiers are opaque, and grounding exposes the full content-region sentence pool rather than a gold-only candidate set. Family scorers are deterministic and frozen before model outcomes; a pre-execution self-test gives 23/23 success on gold outputs and 0/23 on constructed negatives.

A post-review validation adds 12 EIA Weekly Petroleum Status Report cutoff endpoints from a third institutional system (U.S. Energy Information Administration, 2026). EIA is chosen by a structural compatibility rule—recurring first-party releases, explicit publication dates, represented data-ending weeks, and compatibility with the already frozen cutoff interface—before any EIA model outcome is opened. Four endpoints form an earlier diagnostic split and eight April–May endpoints form held-out validation. T1 and G1 are reused byte-for-byte; source manifest, endpoints, scorer, harness, and condition order are frozen before execution. This is informed post-review validation, not an independently preregistered confirmation.

5.2 QUESTIONS AND INFERENCE CONTRACT

RQ1 asks whether adding the no-skill anchor changes a verdict that would look positive under targeted-versus-generic comparison alone. RQ2 asks whether targeted procedures retain a residual over behaviorally neutral extra computation after G_0 is independently validated against fresh N. RQ3 asks whether byte-identical procedures retain direction on compatible held-out releases while preserving ceiling and incompatible-domain nulls. RQ4 asks whether callable skill state adds material value over an operation-matched retrieval/context treatment R that exposes exactly the same frozen temporal-operation output.

The endpoint is the independent scientific unit. Repeats characterize execution robustness and are averaged within endpoint; they are never counted as additional tasks. For non-ceiling cells we report endpoint win/tie/loss patterns, percentile bootstrap intervals over endpoints, and exact two-sided sign tests over non-ties when resolution permits. Two- and three-endpoint cells are explicitly descriptive. We do not pool heterogeneous procedures, models, or domains into an omnibus procedure-effect statistic.

5.3 EVIDENCE LAYERS AND PARITY

DeepSeek-V4-Pro is the registered primary track with five-repeat historical replay plus fresh identification runs. A pre-outcome support rule selects Kimi-K3 as a separately labeled replication; a later hash audit removes one overwritten historical repeat block without replacement. Doubao-Seed-2.0-Mini is an exploratory capacity stress. The EIA layer evaluates DeepSeek and the mini model. After adding the prospectively frozen G_0 and R controls, the retained source-native evidence ledger contains 2,056 runtime-valid rows over the same 35 distinct endpoints from BEA, NOAA, and EIA. The endpoint count, not the enlarged repeat/control-row count, determines scientific resolution.

Every targeted procedure implementation has a family-matched generic helper with the same callable signature, identical Python lexical-token count and AST-node count, and equal external tool-call budget. G_0 uses the same helper-execution surface but is information-empty: it returns $\{\}$ and reads neither package nor context. For R, zero-call parity verifies on every tested endpoint that removing the added retrieval-operation field recovers the original evidence package exactly and that the materialized operation output is content-identical to T’s helper output. Full execution chronology, content hashes, per-call CSV checkpoints, and provenance bindings are reported in Appendix A and the supplement.

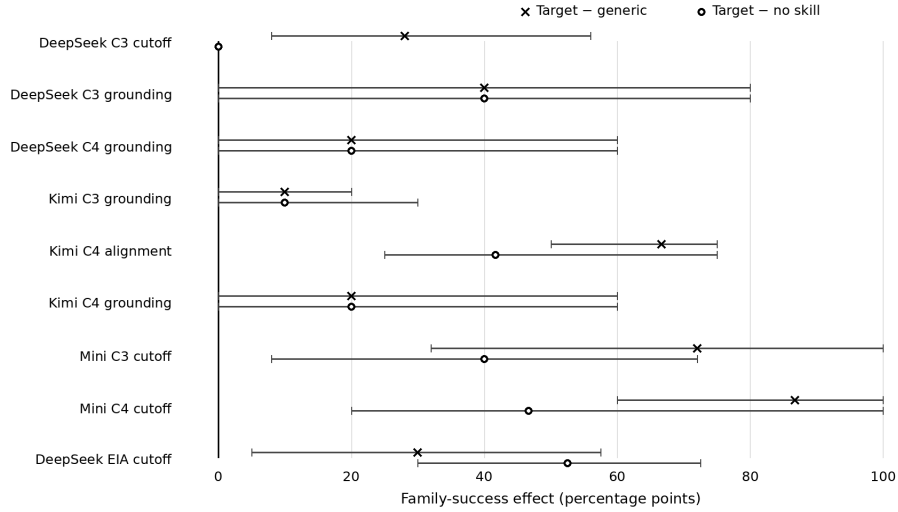


Figure 2: Temporal-procedure effects are regime-dependent. Circles compare targeted with the original no-skill agent; crosses compare targeted with the matched generic helper. Horizontal bars show endpoint-bootstrap 95% intervals. DeepSeek C3 cutoff illustrates the identification point: $T > G$ but $T = N$, so the apparent two-arm gain is not a repair. EIA denotes the frozen post-review energy validation.

6 RESULTS

We organize the results around the three predictions from Section 2: whether the no-skill anchor changes a two-arm verdict, where a targeted procedure is binding rather than ceiling-limited, and whether unchanged procedures retain direction on compatible future releases. Endpoints are the independent units; repeats measure execution robustness only.

6.1 THE THIRD ARM CHANGES THE SCIENTIFIC VERDICT

The clearest justification for the design is a negative result. On DeepSeek C3-R cutoff, the targeted procedure succeeds on 100% of endpoint-repeat observations and the matched generic helper on 72%. A conventional T-versus-G comparison would therefore report a +28-point benefit. The no-skill agent, however, is also at 100%. Thus $\hat{\Delta}_{T-G} = +28$ points but $\hat{\Delta}_{T-N} = 0$, and Definition 1 classifies the cell as *no repair*.

The frozen G1 helper explains why this is a plausible failure mode for a generic control: it validates schema-compatible evidence but is deliberately forbidden from comparing release dates with the cutoff, so it can foreground both an admissible release and a post-cutoff distractor. T1 performs the missing cutoff operation, but in this cell it only restores the behavior of the original agent. The same pattern appears more weakly in DeepSeek C4-R, while Kimi cutoff is at ceiling. The three-arm design therefore does more than reduce variance: it prevents a generic-arm deficit from being promoted into a procedure-repair claim.

A zero-call r9 audit further shows that T-G is not explained by unequal helper availability: targeted and generic helpers are each forced exactly once in all 442 retained rows of their respective arms, versus 0/442 in no-skill; condition position has no detectable success association in any arm ($p > .65$), and targeted prompts are 18.85 input tokens shorter on average than their exact generic counterparts. These diagnostics close simple invocation/order/length explanations for T-G but do not make G behaviorally neutral. We therefore execute the stronger G_0 control below rather than treating these diagnostics as a substitute.

The load-bearing historical cells are DeepSeek C3 grounding (20/20/60 for N/G/T), DeepSeek C4 grounding (40/40/60), and Kimi C4 alignment (58/33/100); the complete matrix, including ceiling/null cells and the later Kimi-grounding downgrade, is reported in Appendix A.

6.2 STRONGER CONTROLS SHARPEN THE IDENTIFICATION BOUNDARY

G_0 is admitted only after independent equivalence to fresh N: the frozen DeepSeek A2 replication gives $G_0-N = -2.1$ points, 90% CI $[-9.3, 5.0]$, and Kimi gives $+2.2$ points, CI $[-4.3, 8.7]$. DeepSeek C3 grounding, C4 grounding, and held-out EIA cutoff then retain positive $T-G_0$ direction; Kimi same-domain grounding is downgraded ($T-G_0 = -25$ points, $T-N=0$), while Kimi alignment retains $+50$ points over both fresh controls.

R resolves the container question. On 18 pre-frozen DeepSeek load-bearing endpoints, $T-R$ is 0.0 points with 90% CI $[-5.6, 5.6]$ and no positive cell residual; three held-out Kimi alignment endpoints likewise give $T=R=100\%$. Thus the data support the operation, not a privileged callable container. Full staging and parity audits are in Appendix A.

6.3 GROUNDING IS A BINDING REPAIR WHEN THE ORIGINAL AGENT HAS HEADROOM

DeepSeek gives the cleanest grounding case: C3-R is 20%/20%/60% for N/G/T (+40 points over both controls), and C4-R is 40%/40%/60%. The fresh neutral control preserves positive direction on both pre-frozen load-bearing cells. The effect is not cross-model universal: Kimi same-domain grounding is null after G_0 , and cross-domain grounding is also null. We therefore retain grounding as a DeepSeek-regime result rather than a general transfer claim.

6.4 CUTOFF REPAIR APPEARS IN NON-CEILING REGIMES AND DISAPPEARS AT CEILING

Cutoff effects expose the same regime boundary. The exploratory mini model has headroom on the initial replay, whereas DeepSeek C3-R is already at N ceiling. The stronger held-out EIA test freezes 12 new energy endpoints and reuses T1/G1 byte-for-byte: on eight held-out endpoints DeepSeek moves from 47.5% (N) and 70% (G) to 100% (T). $T-N$ is positive on seven of eight endpoint means (exact sign test $p = 0.0156$, 95% bootstrap interval $[30, 72.5]$ points). The mini model is instead 40/40 under N on the same EIA endpoints, so the cutoff operation is binding only when the model-task cell has headroom; the full exploratory matrix is in Appendix A.

6.5 TRANSFER IS SELECTIVE AND EXPLICITLY BOUNDED

Held-out Kimi alignment moves away from ceiling historically ($T/N/G=12/12, 7/12, 4/12$), and the fresh control preserves $T-G_0=T-N=+50$ points on the same three endpoints. Yet operation-matched R is equally effective: T and R both succeed on all five fresh repeats. Combined with the Kimi-grounding downgrade, cross-domain grounding null, and EIA ceiling cells, transfer is selective and does not imply a uniquely effective callable surface.

7 LIMITATIONS

The unavailable TimeSage-EV evaluated snapshot prevents an exact rerun, so our 35-endpoint BEA/NOAA/EIA replay is independent mechanistic evidence with uneven resolution: EIA is cutoff-focused, same-domain grounding transfer has two held-out endpoints, Kimi alignment has three, and cross-domain grounding is null. DeepSeek is primary, Kimi a replication, and the mini model exploratory; repeats measure robustness rather than task sample size. G_0 establishes neutrality only under the frozen ± 10 -point rule, and R rules out a material callable-container advantage only for the tested same-information integration surface. We therefore do not claim broad cross-domain/model-invariant transfer, neutrality of the historical G helper, or equivalence to every learned temporal retriever, indexing policy, or context budget.

8 CONCLUSION

Reusable temporal procedures can be binding repairs in specific non-ceiling model-task regimes. Requiring T to beat the original agent turns one apparent $+28$ -point targeted-versus-generic cutoff effect into no repair; validating G_0 then separates operation value from generic-helper harm; and operation-matched R shows that the same repair need not live in a callable skill container. The strongest supported conclusion is therefore not “skills help.” It is that recurring temporal *operations*

can become causal bottlenecks, and those operations should be credited only after original-agent, neutral-compute, and same-information integration controls have been made explicit. In our tested regime, retrieval/context construction is as effective as callable skill state for carrying the operation.

REPRODUCIBILITY STATEMENT

The source-native study is content-addressed end to end. The supplementary replication bundle contains the frozen endpoint definitions, prompt harnesses, family scorers, all six executed targeted/generic helper implementations, condition schedules, first-party archived source pages, retained raw result JSON, adjudication code, and a SHA-256 manifest. Appendix A records the executed helper code, the endpoint-level inferential protocol, and the zero-call invocation/order/date-serialization audits. The source-native bundle is independent of the unavailable TimeSage-EV evaluated snapshot; the exact TimeSage replication remains a separately identified external replication target.

AI USE STATEMENT

Generative AI tools were used throughout the research workflow to assist literature triage, research-idea and hypothesis refinement, experimental-design critique, implementation and debugging, data reformatting, result interpretation, internal mock review, and manuscript drafting and editing. AI-assisted experimental artifacts were accepted only after executable checks against frozen task contracts, immutable raw outputs, first-party source material, or deterministic recomputation. AI-generated review feedback had advisory status and did not itself create scientific evidence. The authors reviewed the AI-assisted work and take responsibility for the final paper, claims, code, and supplementary artifacts.

REFERENCES

- Abdelrahman Abdallah, Bhawna Piryani, Jonas Wallat, Avishek Anand, and Adam Jatowt. Tempree-triever: Fusion-based temporal dense passage retrieval for time-sensitive questions. In *Proceedings of the 19th ACM International Conference on Web Search and Data Mining*, pp. 5–15. Association for Computing Machinery, 2026. doi: 10.1145/3773966.3777938.
- Akari Asai, Zeqiu Wu, Yizhong Wang, et al. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024.
- Gen Dong, Yanjie Gao, Liqun Li, et al. Agent skills can be harmful: An empirical study of skill-induced failures in llm agents, 2026.
- Tianyi Guan, Yiding Wang, Haotong Yang, et al. Continualskillbench: Can llm agents truly evolve their capabilities?, 2026.
- Jiale Han, Austin Cheung, Yubai Wei, et al. Rag meets temporal graphs: Time-sensitive modeling and retrieval for evolving knowledge, 2025.
- Hanbing Liu, Lang Cao, and Yang Li. RAG or learning? understanding the limits of LLM adaptation under continuous knowledge drift in the real world. In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 11234–11252. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.546. URL <https://aclanthology.org/2026.findings-acl.546/>.
- NOAA Climate Prediction Center. ENSO Diagnostic Discussion Archive. https://www.cpc.ncep.noaa.gov/products/expert_assessment/ENSO_DD_archive.php, 2026. First-party monthly discussions accessed 2026-08-22.
- Qianjun Pan, Yutao Yang, Junsong Li, Jie Zhou, Kai Chen, Xin Li, Qin Chen, and Liang He. Anything2skill: Compiling external knowledge into reusable skills for agents, 2026.
- U.S. Bureau of Economic Analysis. GDP News Release Archive. <https://www.bea.gov/news/archive>, 2026. First-party historical releases accessed 2026-08-22.

U.S. Energy Information Administration. Weekly Petroleum Status Report Archive. <https://www.eia.gov/petroleum/supply/weekly/archive/>, 2026. First-party weekly release archive accessed 2026-08-22.

Xiaoman Wang, Yaoze Zhang, Wenzhuo Fan, et al. Ia-rag: Interval-algebra-driven temporal reasoning for dynamic knowledge retrieval, 2026a.

Yixuan Wang, Yiyang Zhou, Yiming Liang, et al. Not all skills help: Measuring and repairing agent knowledge, 2026b.

Tianxin Wei, Zhan Shi, Minhua Lin, et al. Evo-harness: Context-to-harness skill compilation for self-evolving agents, 2026.

Ruiyao Xu, Tiankai Yang, and Wei-Chieh Huang. Hyperskill: Self-evolving llm agents via hypergraph-structured skill memory, 2026.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation, 2024.

Qingren Yao, Yaxuan Kong, Yuqi Nie, Yichen Li, Anna Vettoruzzo, Qingsong Wen, Ming Jin, Joaquin Vanschoren, and Stefan Zohren. Timesage-ev: A live benchmark for agentic time series analysis in evolving environments, 2026.

Siyue Zhang, Yuxiang Xue, Yiming Zhang, Xiaobao Wu, Anh Tuan Luu, and Chen Zhao. MRAG: A modular retrieval framework for time-sensitive question answering. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 3080–3118. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-emnlp.167. URL <https://aclanthology.org/2025.findings-emnlp.167/>.

Mingxuan Zheng, Yujin Zhou, Chuxue Cao, et al. Skillprox: Self-evolving agent skills via proximal textual gradient descent, 2026.

Xiaolin Zhou, Jinbo Liu, Li Li, et al. Counterfactual trace auditing of llm agent skills, 2026.

A REPRODUCIBILITY AND SENSITIVITY AUDITS

T1/G1 are byte-identical across the initial replay and EIA validation. The provenance-clean r3 reaudit is 6221016166b71ebe; r4 adjudication is 316ee89a8eb1d53f. The r5 replication bundle contains executable r3/r4 endpoints, harnesses, scorers, runner scripts, frozen helper source, first-party archived pages, retained result JSON, and a SHA-256 manifest. This bundle reproduces the source-native study and remains separate from the unavailable TimeSage-EV evaluated snapshot.

A.1 EXACT FROZEN REUSABLE HELPERS

The six helper implementations below are the executed source, not pseudocode. Each targeted/generic pair has equal lexical-token count and equal AST-node count under the pre-execution audit.

```
# T1_temporal_cutoff.py
def skill(package, context):
    limit = context.get("cutoff_date", "")
    chosen = []
    for item in package.get("evidence_metadata", []):
        marker = item.get("release_date", "")
        ref = item.get("evidence_ref", "")
        if isinstance(marker, str) and marker <= limit:
            chosen.append(ref)
    return {"evidence_refs": chosen}

# G1_temporal_cutoff.py
def skill(package, context):
    limit = context.get("schema_ceiling", "")
    chosen = []
    for item in package.get("evidence_metadata", []):
        marker = item.get("schema_version", "")
        ref = item.get("evidence_ref", "")
```

```

594         if isinstance(marker, str) and marker <= limit:
595             chosen.append(ref)
596         return {"evidence_refs": chosen}

597 # T2_release_alignment.py
598 import re
599 def skill(package, context):
600     text = " ".join(package.get("document_text", "").split())
601     match = re.search(r"([1-4]) (?:(st|nd|rd|th) Quarter(?: and Year)? (20\d{2}))",
602                     text, re.I)
603     value = None
604     if match:
605         value = f"{match.group(2)}-Q{match.group(1)}"
606     return {"value": value}

607 # G2_release_alignment.py
608 import re
609 def skill(package, context):
610     text = " ".join(package.get("document_text", "").split())
611     match = re.search(r"(News) (Release|Archive|Statement) (?: and Data)? "
612                     r"([A-Za-z]{2,20})", text, re.I)
613     value = None
614     if match:
615         value = f"{match.group(1)}-{match.group(3)}"
616     return {"value": value}

617 # T3_exogenous_grounding.py
618 def skill(package, context):
619     markers = ("contributors to", "reflected", "because", "collectively",
620              "favored", "favors ", "associated with", "due to")
621     chosen = []
622     for span in package.get("candidate_spans", []):
623         text = span.get("text", "").lower()
624         sid = span.get("span_id", "")
625         if any(marker in text for marker in markers):
626             chosen.append(sid)
627     return {"span_ids": chosen}

628 # G3_exogenous_grounding.py
629 def skill(package, context):
630     markers = ("table", "figure", "release", "appendix", "contact",
631              "notes ", "technical notes", "page")
632     chosen = []
633     for span in package.get("candidate_spans", []):
634         text = span.get("text", "").lower()
635         sid = span.get("span_id", "")
636         if any(marker in text for marker in markers):
637             chosen.append(sid)
638     return {"span_ids": chosen}

```

A.2 EXPLORATORY CAPACITY-STRESS MATRIX

Split	Family	N0	Generic	Target	Δ_{T-G}	Δ_{T-N}
C3-R	Cutoff	32	0	72	+72	+40
C3-R	Align	100	100	100	0	0
C3-R	Ground	40	48	64	+16	+24
C4-R	Cutoff	53	13	100	+87	+47
C4-R	Align	80	93	100	+7	+20
C4-R	Ground	60	52	64	+12	+4

Table 1: Exploratory mini-model capacity stress over five repeats (percent), reported separately from the registered primary and replication tracks.

A.3 ENDPOINT-LEVEL UNCERTAINTY AND TEST RESOLUTION

Table 2 averages repeats within endpoint-condition and bootstraps endpoints; W/T/L is targeted versus generic, and sign tests use non-ties. Repeats never increase the inferential sample size. For n non-tied endpoints, the smallest attainable two-sided sign-test value is 2^{1-n} when every endpoint moves in the same direction. Thus two non-tied endpoints cannot yield a value below .5; three cannot go below .25; five cannot go below .0625; eight can reach .0078. Ties reduce resolution further. This arithmetic is why the two-endpoint grounding-transfer result is explicitly descriptive, why the

three-endpoint Kimi alignment result remains underpowered despite a large effect, and why the eight-endpoint EIA validation supports a stronger endpoint-level claim.

Layer	Split	Fam.	Δ_{T-G} [95%]	W/T/L	p_G	Δ_{T-N} [95%]	p_N
D	C3	Cut	28 [8,56]	4/1/0	.125	0 [0,0]	1
D	C3	Grd	40 [0,80]	2/3/0	.50	40 [0,80]	.50
D	C4	Cut	13 [0,20]	2/1/0	.50	0 [0,0]	1
D	C4	Grd	20 [0,60]	1/4/0	1	20 [0,60]	1
K	C3	Grd	10 [0,20]	2/3/0	.50	10 [0,30]	1
K	C4	Align	66.7 [50,75]	3/0/0	.25	41.7 [25,75]	.25
K	C4	Grd	20 [0,60]	1/4/0	1	20 [0,60]	1
M	C3	Cut	72 [32,100]	4/1/0	.125	40 [8,72]	.25
M	C3	Grd	16 [-12,60]	1/3/1	1	24 [0,64]	.50
M	C4	Cut	86.7 [60,100]	3/0/0	.25	46.7 [20,100]	.25
M	C4	Align	6.7 [0,20]	1/2/0	1	20 [0,60]	1
M	C4	Grd	12 [0,28]	2/3/0	.50	4 [-32,36]	1

Table 2: Endpoint-level uncertainty for non-ceiling initial-replay cells. D=DeepSeek primary, K=Kimi replication, M=exploratory mini.

A.4 WHY THE FAMILY ESTIMAND IS PRIMARY

The family contract and the full-task conjunctive score answer different questions. The family score asks whether the targeted temporal operation was corrected. The conjunctive score additionally requires orthogonal output fields and exact surface forms. A zero conjunctive score can therefore coexist with a successful cutoff intervention when temporal admissibility is correct but an unrelated field differs, or with successful release alignment when a semantically identical period is formatted differently. The family definitions were frozen before execution, so selecting them is determined by the experimental question and its time order. The conjunctive score is preserved as a sensitivity view and is never used to erase an unfavorable family result.

Family	r3 full-task conjunctive score	Effect on claim
Grounding	Identical to family score	Unchanged
Cutoff	All cells collapse to 0	Family estimand remains primary
Alignment	All cells collapse to 0	Family estimand remains primary

Table 3: Sensitivity to the auxiliary full-task exact score. The two columns evaluate different estimands.

A.5 CALL-RATE, PROMPT-LENGTH, AND CONDITION-POSITION AUDIT

The retained source-native rows contain 442 no-skill, 442 generic, and 442 targeted observations. Helper execution is deterministic in the frozen harness: generic and targeted execute exactly once before the model answer in 442/442 rows each, whereas no-skill executes no helper in 0/442 rows. Thus targeted-minus-generic cannot be explained by a higher helper invocation rate; this experiment does not measure endogenous tool-use propensity. Across exact targeted/generic pairs, targeted prompts are shorter by 18.85 input tokens on average (302 shorter, 140 longer). Condition-position counts are 157/135/150 for no-skill, 145/155/142 for generic, and 140/152/150 for targeted across positions 0/1/2. Pooled 3×2 success-by-position tests give $p = .993$, $.657$, and $.914$ for no-skill, generic, and targeted respectively; targeted-minus-no-skill remains positive at all three positions. These are post-hoc zero-call diagnostics and do not substitute for a prospectively counterbalanced replication. The full audit is included in the supplement.

A.6 ISO DATE AND PERIOD-REPRESENTATION AUDIT

A zero-call static audit covers the 23 initial BEA/NOAA endpoints and the 12 post-review EIA endpoints. Across the serialized endpoint objects, all 55 occurrences of `cutoff_date` and all 176 occurrences of `release_date` are calendar-valid YYYY-MM-DD strings. Whenever both top-level and `skill.context` cutoff fields are present they agree exactly. T1 makes 88 observed release-versus-cutoff comparisons over the frozen endpoint packages; lexical string order and parsed-calendar order agree on all 88. The executed T1 source is byte-identical between the initial and EIA phases (SHA-256 `df1950ef3930...d56d522`). Reporting-period identities are a separate

representation layer: the initial replay uses canonical month/quarter identities for relevant families, while EIA uses day-level data-ending dates. These reporting-period strings never enter T1’s release-date comparator and are adjudicated by the frozen family scorer.

A.7 COMPLETED IDENTIFICATION LADDER

Table 4 separates four distinct questions that are all executed on the source-native replay. The controls were added prospectively in sequence: outcome thresholds, endpoint units, integration surfaces, model identities, and checkpoint policies were frozen before the corresponding fresh outcomes.

Question	Contrast	Status	Identified quantity
Repair original agent?	T-N	Executed	Incremental value of inserting the targeted procedure over the original agent.
Beat target-blind help?	T-G	Executed	Targeted-operation value relative to the frozen complexity-matched helper.
Beat neutral extra help?	T-G ₀	Executed	Operation-specific value after G ₀ independently passes a fresh N-equivalence gate.
Does container matter?	T-R	Executed	Callable-skill value beyond the same frozen operation materialized in retrieved context.

Table 4: Completed identification ladder. The stronger G₀ and R treatments change the interpretation of the historical N/G/T results: G₀ validates operation specificity in surviving cells, whereas R removes any supported claim of callable-container superiority.

G₀: prospectively validated neutral helper. G₀ executes exactly one assigned helper call through the same harness surface as T but is information-empty: the frozen function is `def skill(package, context): return {}`. It reads neither package nor context. On DeepSeek, the first 35-endpoint two-repeat tranche is deliberately not promoted because its G₀-N 90% interval $[-14.3, 2.9]$ points exceeds the predeclared ± 10 -point neutrality margin. We do not pool that tranche or relax the margin. An independently frozen A2 replication with four fresh repeats yields -2.1 points, 90% CI $[-9.3, 5.0]$, and passes the standalone equivalence gate. Kimi independently gives $+2.2$ points, 90% CI $[-4.3, 8.7]$. Only after these gates do we inspect the pre-frozen T-G₀ load-bearing cells. DeepSeek C3 grounding, C4 grounding, and held-out EIA cutoff retain positive direction; Kimi alignment retains $+50$ points, while Kimi same-domain grounding is downgraded (T-G₀ = -25 points, T-N=0).

R: operation-matched retrieval/context treatment. R runs the exact same targeted operation source as T on the same candidate evidence. Rather than exposing the output through `reusable_helper_output`, the harness adds the identical object as `retrieval_operation_output` inside the evidence package and exposes no callable helper. A zero-call parity audit verifies for every tested endpoint that (i) removing this one field recovers the original evidence package exactly, (ii) the operation-output content hashes are identical between T and R, and (iii) no candidate evidence is removed or reordered. On all 18 pre-frozen DeepSeek load-bearing endpoints, two fresh repeats per arm give T-R = 0.0 points with 90% CI $[-5.6, 5.6]$; all three load-bearing cells have zero mean residual. A secondary Kimi alignment check uses the three pre-frozen held-out alignment endpoints and five fresh repeats per arm: both T and R succeed on every run. We therefore rule out a material > 10 -point callable-container advantage at the registered primary resolution. This is a same-information integration result, not an evaluation of every possible learned temporal retriever.

Checkpointed execution and recovery. Every fresh G₀ and R model call is persisted before the next unit begins: an atomic raw JSON receipt, one fsynced CSV row, one fsynced JSONL row, and an atomic checkpoint summary keyed by `endpoint×repeat×arm`. Resume logic executes missing units only and never reruns a successfully checkpointed unit. Each experiment first runs a small runtime/protocol pilot whose promotion rule ignores scientific outcomes. This design was exercised in practice: transient control-channel failures occurred during the Kimi run while the remote process continued, and the already completed rows remained recoverable from CSV/raw checkpoints without replaying prior calls.

756 Every retained row binds endpoint, condition, repeat, skill hash, prompt hash, raw response hash, and
757 provider response-ID hash. Kimi repeat 4 is excluded by hash provenance with no replacement. The
758 exact TimeSage-EV replication remains separate from all source-native evidence.
759
760
761
762
763
764
765
766
767
768
769
770
771
772
773
774
775
776
777
778
779
780
781
782
783
784
785
786
787
788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809